

Diagnostic questions

Working scientifically

1. Students investigate the physical properties of some substances. The table below shows how the different materials have been grouped.

What are the appropriate labels for each column?

Label 1	Label 2	Label 3
Aluminium foil Brass key Gold ring	Cooking oil Soap bubble wood chip feather	Baking soda Squash concentrate sugar

A Good conductor, more dense than water, soluble in water

B Good conductor, less dense than water, solvents

C Good conductor, less dense than water, soluble in water

D Insulator, more dense than water, soluble in water

E Insulator, less dense than water, solvent

Answer C

2

A class of 8 students carried out an experiment to find out how much energy was contained in a biscuit.

Method:

1. Burn a piece of biscuit under a test tube of water
2. Record the temperature change

Results:

Student	1	2	3	4	5	6	7	8
Temperature change (°C)	13	53	23	3	42	17	29	25

Using the information above, is this method reproducible?

A - Yes because all the students got a result

B - Yes because the 8 students can calculate a mean of all the results

C - No because they didn't repeat it three times and calculate the mean

D - No the results are not similar to each other because the method is not detailed

Answer D

3.

How is 1 nanometre written in standard form?

- A. $1 \times 10^{-9} \text{ m}$
- B. $1 \times 10^{-6} \text{ m}$
- C. $1 \times 10^{-3} \text{ m}$
- D. 0.000000001 m

Answer A

4.

The results table below shows how fast three different colour of paint diffused in 1 litre of water.

Paint colour	Time taken to diffuse in 1 litre of water (min)
Red	2
Purple	3
Black	6

What is the independent variable in this investigation?

- A – The different colours of paint
- B – The black paint took longest to diffuse
- C – The time taken to diffuse in 1 litre of water (min)
- D – Measuring how fast the colours took to diffuse

Answer A

5.

Answer A

The mass of a potato increased from 2g to 2.5g when placed in water. What is the percentage increase?

A. 25%

B. 125%

C. 2%

D. 20%

6. Richard wanted to find out the best conditions for growing lettuce plants.



He took 4 trays and planted 8 lettuce plants in each.
The results of his investigation are shown below.

tray	variables			number of plants alive after 7 days
	light level	air temperature (°C)	soil moisture	
A	medium	25	moist	8
B	medium	25	dry	6
C	medium	45	moist	2
D	medium	45	dry	0

What conclusions can Richard make from his results?

- A. All variables affect the number of plants alive after 7 days
- B. Only the temperature affect the number of plants alive after 7 days
- C. Only soil moisture affect the number of plants alive after 7 days
- D. Both temperature and soil moisture affect the number of plants alive after 7 days
- E. He can't make any conclusions as he had more than 1 independent variable.

Answer D.

7.

What is accuracy?

A

Getting a precise result

B

Getting a valid result.

C

Getting a result as close as possible to the true value

D

Getting the same result on several occasions.

Answer C

8.

A scientist has invented a new material called Staywarm for coats which is a good insulator.

4 materials are tested. One of the materials is Staywarm.

Which material would you predict is Staywarm?

(b) The results of the investigation are shown below.

time (minutes)	temperature of water (°C) wrapped in			
	material A	material B	material C	material D
0	60	60	60	60
20	34	40	38	36

A

B

C

D

Answer B

9.

A pupil is investigating how strong his concrete bridge is. He adds 10g masses until the bridge breaks. Another student says he could repeat this but be more accurate. How?

A

Repeat the experiment using more bridges and calculate a mean strength.

B

Time how long it takes for the bridge to break after adding each mass.

C

Add 1g masses instead of ten

D

None of these improve the accuracy

Answer C

10.

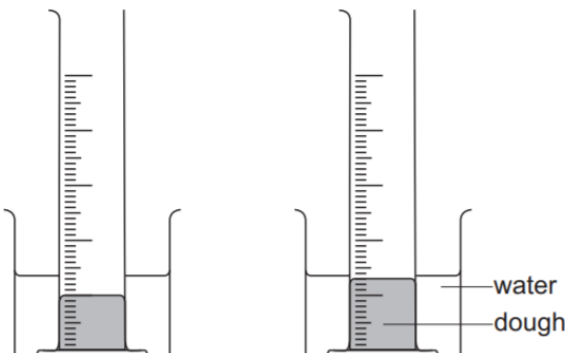
4. Sara investigated making bread.
She described what she did below.

I mixed flour, water, sugar and yeast to make bread dough.

I put 50 cm³ of dough into a measuring cylinder.

I put the measuring cylinder into a water bath at 30°C.

I measured the volume of the dough after 30 minutes.



at the start after 30 minutes

Sara repeated the experiment with the water bath at different temperatures.
Her results are shown below.

temperature of water bath (°C)	volume of dough (cm ³)	
	at the start	after 30 minutes
30	50	66
45	50	73
60	50	77
75	50	71
90	50	60

Which question did Sara investigate?

- A. By how much can I make dough rise?
- B. What happens to the dough after 30 minutes?
- C. Which temperature causes the dough to rise the most?
- D. How does temperature affect how much the dough rises?

Answer D